conditions like Pityriasis versicolor, Seborrheic dermatitis, Atopic dermatitis, Malassezia folliculitis and systemic infection. Symptoms include macular, erythematous, hyper pigmented or hypo pigmented lesions with fine scaling.

Tinea nigra is responsible for the superficial cutaneous infection of the skin. *Hortaea werneckii* is the phaeoid (dematiaceous) fungi causes infection on the palms and soles. It is also commonly termed as *Tinea nigra palmaris* and *Tinea nigra plantaris*. Symptoms includes brown to black deeply pigmented non - scaly, macular lesions affecting skin of the palms and occasionally soles.

Piedra causes superficial infection of hair shaft. The word **Piedra** is derived from Spanish word **Stone**. There are two types of Piedra based on causative fungi and characteristics of nodules. They are **Black piedra** caused by *Piedraia hortae* and **White piedra** caused by *Trichosporon* species. The symptoms include development of firm, irregular nodules of fungal elements cemented to the hair. The piedra can be distinguished on the basis of shape, size and pigmentation of fungal cells of nodules which are found around hair cortex.

9.2.1 Dermatophytoses

Dermatophytoses are the most common cutaneous fungal infection seen in man and animals affecting skin, hair and nails. The fungi can invade the keratinized tissues of skin and its appendages and they are collectively known as **Dermatophytes** or **Tinea** or **ring worm** infection. The dermatophytes are hyaline septate molds. They are divided into three main anamorphic genera depending on their morphological characteristics.

i. ***Trichophyton*** [Cause infection in skin, hair and nails]

ii. ***Microsporum*** [Cause infection skin and hair]

iii. ***Epidermophyton*** [cause infection skin and nail]

The fungal species affecting humans are known as **anthropophilic**. Those inhabitating domestic and wild animals as well as birds are called **zoophilic**. Fungi species from soil are known as **geophilic** dermatophytes.

HOTS

What are the sources of dermatophytes?

9.2.2 Pathogenesis and Pathology

The dermatophytes grow within dead keratinized tissue and produce keratinolytic proteases, which provide means of entry into living cells. Fungal metabolic products cause erythema, vesicles and pustule on the site of infection. Some dermatophytes species like soil saprobes digest the keratinaceous debris in soil and are capable of parasitizing keratinous tissues of animals.

9.2.3 Clinical Features

The clinical manifestations of Dermatophytoses are also called **Tinea or Ringworm** depending on the anatomical site involved. Following are the common clinical conditions produced by dermatophytes:

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